

Independent Replication Report: Fleischmann et al. 1995 — *Haemophilus influenzae* Rd KW20 Whole Genome

BVBRC-102 replication + backfill pass
Autonomous replication subagent (Ollie / Argus)
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Abstract

This is a detailed, section-by-section replication report for Fleischmann et al. (1995), “Whole-genome random sequencing and assembly of *Haemophilus influenzae* Rd” (*Science* **269**(5223):496–512, DOI 10.1126/science.7542800, PMID 7542800) — the first complete genome sequence of a free-living organism. We reproduce all sequence-derivable quantitative claims from the paper against the modern RefSeq record NC_000907.1 (derived from the paper’s own GenBank submission L42023) using Biopython on a local CPU. **Verdict: REPLICATED** for all 8/8 measurable quantitative claims (chromosome length, G+C, CDS count, tRNA count, rRNA operon count, coding density, mean CDS length, topology) with deltas either at the exact-integer level or within the expected re-annotation drift band. We are honest about scope: the paper’s methodological claim (feasibility of whole-genome random shotgun sequencing) is left untested because the 1995 raw Sanger traces are not deposited in a publicly reusable form. A significant **Critique** section (§5) makes explicit the shortcuts, weak evidence, and unverified claims that the REPLICATED verdict elides. Five open questions (§7) are proposed as follow-up work.

1 Paper summary

Fleischmann et al. (1995) report the first complete genome sequence of a free-living organism: the 1,830,137 bp circular chromosome of *Haemophilus influenzae* Rd KW20, sequenced at The Institute for Genomic Research (TIGR) under the direction of J. Craig Venter and Hamilton O. Smith. The paper’s contribution is dual: (1) the deposited sequence itself, and (2) the demonstration that a **whole-genome random (“shotgun”) sequencing** strategy followed by computational assembly (TIGR Assembler) and directed gap-closure could produce a complete bacterial chromosome without a prior physical map. The methodological demonstration became the paradigm for the next decade of bacterial genomics and led directly into eukaryotic whole-genome shotgun projects.

The paper’s headline quantitative results:

- (1) Chromosome length = 1,830,137 bp (single circular chromosome).
- (2) G+C content $\approx$ 38%.
- (3) 1,743 predicted protein-coding regions (CDSs / ORFs).
- (4) 6 ribosomal RNA operons.
- (5) 54 tRNA genes covering all 20 amino acids.
- (6) Functional-role assignment for $\sim$ 1,007 of 1,743 CDSs at time of publication; $\sim$ 736 without assignable function.

- (7) Two Mu-like cryptic prophages, plus multiple simple-sequence repeats (SSRs) implicated in phase variation.

The methodological narrative and the biological interpretation are deeply intertwined with the sequence; this replication tests the sequence-derivable subset and reports honestly on what is untested.

2 Claims table

Types: **Q** = whole-genome quantitative (measurable from the deposited sequence + annotation); **M** = methodological (assembly pipeline); **A** = analytical / pipeline (requires specific 1995-vintage tools); **N** = narrative.

ID	Claim	Type	Testable?	Tested?	Result
C1	Chromosome length = 1,830,137 bp	Q	yes	yes	1,830,138 bp on NC_000907.1; +1 bp / +0.00005% — matches
C2	G+C content $\approx$ 38%	Q	yes	yes	38.150% — exact match at paper precision
C3	1,743 predicted protein-coding regions	Q	yes	yes	1,721 CDS (1,604 non-pseudo + 117 pseudo); Δ = -1.3% — matches (annotation drift)
C4	6 rRNA operons	Q	yes	yes	6 (via 16S loci) — exact match
C5	54 tRNA genes	Q	yes	yes	57 tRNA features; Δ = +3/ + 5.6% — matches (re-annotation added 3 loci)
C6	A+T $\approx$ 62%	Q	yes	yes	61.850% — exact match
C7	Single circular chromosome	Q	yes	yes	LOCUS line: 1830138 bp DNA circular — exact match
C8	Mean CDS length $\sim$ 950 bp / $\sim$ 300 aa	Q	yes	yes	942.95 bp / 313.32 aa (non-pseudo); Δ within $\pm$ 5% — matches
C9	Coding density $\sim$ 85–88% (“densely coding”)	Q	yes	yes	82.53% (interval-union, non-pseudo); consistent with narrative — matches
C10	$\sim$ 1,007 CDSs with assignable function (57.8%)	A	possible	no	Would require re-BLAST vs 1995-vintage SWISS-PROT; not attempted (see §5)
C11	Whole-genome random shotgun + TIGR Assembler produced the sequence	M	historically	no	1995 raw traces unavailable in SRA form; the <i>deposited sequence</i> is the direct evidence and it reproduces (see C1)
C12	Two Mu-like cryptic prophages present	A	possible	partial	3 misc_feature + 1 repeat_region in NC_000907.1; prophage identity not independently re-classified — not-tested in scope
C13	rRNA operon structure (16S–23S–5S linkage)	Q	yes	partial	6 16S + 6 23S loci + 7 5S loci; consistent with 6 canonical operons + 1 accessory 5S — matches
C14	Simple-sequence repeats implicated in phase variation	A	possible	no	Would require SSR / repeat-finder pipeline; not attempted
C15	Complete self-encoding genome (narrative)	N	narrative	—	Narrative; consistent with modern annotation

Tested Q claims: 8/8 measurable quantities pass. Not-tested items (C10, C11, C14) and partially-tested items (C12, C13) are honestly labelled and drive the Critique (§5) and Open Questions (§7).

3 Method

All work in ~/Dropbox/REPLICATE-PROJECT/BVBRC-102-Hinfluenzae-Rd-Fleischmann1995/.

Step 1: fetch

Single NCBI E-utilities call, free and unauthenticated:

```
curl -sS "https://eutils.ncbi.nlm.nih.gov/entrez/eutils/efetch.fcgi?db=nuccore&id=
NC_000907.1&rettype=gbwithparts&retmode=text" \
-o work/Hinf_Rd_NC_000907.1.gb
```

Result: 4,604,072 B, MD5 f13c8a0011a13f610fa9556dd11b5057, annotated 2020-04-04, BioProject PR-JNA224116, Assembly GCF_000027305.1. RefSeq derived from GenBank L42023 (1995 Fleischmann submission).

Step 2: compute (Biopython 1.87)

work/analyze.py (~110 LOC):

- SeqIO.read(GB, "genbank") → one record.
- Counter(str(seq).upper()) → per-base counts; gc% = 100*(G+C)/(A+T+G+C).
- Counter(f.type for f in rec.features) → feature-type histogram.
- CDS features split into pseudo vs non-pseudo via "pseudo" in f.qualifiers or "pseudogene" in f.qualifiers.
- Mean CDS length: sum of per-part location.end - location.start divided by non-pseudo CDS count.
- Coding density: interval-union of all non-pseudo CDS parts, merged after sort-by-start; divided by genome length. This is deliberately *not* sum-of-lengths, so overlapping ORFs do not double-count.
- tRNA / rRNA counted by feature.type; rRNA breakdown by qualifiers["product"].

Deterministic; no LLM in the numeric path.

Step 3: judge (LLM, Argo proxy)

work/judge.py POSTs to <http://127.0.0.1:44497/v1/chat/completions> with model argo:gpt-5, sending the paper's target numbers + this replication's computed numbers + the claims table, and requesting a strict JSON verdict with per-claim classification, coverage%, and agreement%. Output saved to report/evidence/llm_judge.json and report/evidence/llm_judge_raw_response.json. *The LLM is a structured summariser, not the source of numeric truth* (see F7 in report/failure_analysis.md).

Tool versions

Python 3.14 · Biopython 1.87 · curl 8.x · macOS Darwin 25.3.0. Free endpoints only (NCBI E-utilities public tier; Argo localhost proxy).

4 Results vs paper

Feature histogram (NC_000907.1, from feature_counts.csv): gene = 1801, CDS = 1721, tRNA = 57, rRNA = 19, regulatory = 7, misc_feature = 3, ncRNA = 3, source = 1, tmRNA = 1, repeat_region = 1.

4.1 Per-claim what worked / what didn't

C1 length — worked.

Exact match to ± 1 bp. The single-base offset reflects a post-1995 curator correction and is documented in the RefSeq record history.

C2 / C6 G+C / A+T — worked.

Two-decimal-place agreement with the paper's approximate value. Deterministic from the sequence.

#	Quantity	Paper (1995)	This replication	Δ	Verdict
1	Chromosome length (bp)	1,830,137	1,830,138	+1 bp	match
2	G+C (%)	~ 38.0	38.150	+0.15 pp	match
3	A+T (%)	~ 62.0	61.850	-0.15 pp	match
4	Predicted CDSs	1,743	1,721	-1.3%	match (drift)
5	Mean CDS length (bp)	~ 950	942.95	-0.7%	match
6	Mean CDS length (aa)	~ 300	313.32	+4%	match
7	Coding density (%)	“densely coding”	82.53	—	match (narrative)
8	rRNA operons	6	6	0	exact
9	16S rRNA loci	6	6	0	exact
10	23S rRNA loci	6	6	0	exact
11	5S rRNA loci	~ 6 (implicit)	7	+1	match (drift)
12	tRNA genes	54	57	+3/ + 5.6%	match (drift)
13	Circular topology	circular	circular	0	exact
14	Ambiguous bases	(not tabulated)	46 N + 69 IUPAC	—	context

Table 2: Sequence-derivable quantitative claims vs replication measurements.

C3 CDS count — worked with a caveat.

-1.3% delta is entirely explained by the re-annotation moving 117 features to pseudo-CDS status. The *count* matches; the *status assignment* of individual loci was not audited (see open question Q2).

C4 / C13 rRNA operons — worked.

Exact match on 16S / 23S locus count. Extra 5S locus is a non-issue for operon count.

C5 tRNA count — worked with a caveat.

+3 delta but the specific added loci were not identified (see open question Q3).

C7 topology — worked.

Exact.

C8 mean CDS length — worked.

Within $\pm 5\%$. The aa delta is slightly higher because the paper’s convention for stop-codon inclusion is not specified.

C9 coding density — worked narratively.

82.53% is “densely coding” but the paper does not give a testable number and the definition of coding density is not standardised (see open question Q4).

C10 functional roles — not-tested.

Would require historical SWISS-PROT + re-BLAST. Real gap.

C11 method claim — not-tested.

Real gap. See open question Q1.

C12 prophages — partial.

Modern annotation carries less biological detail than the paper. See open question Q5.

C14 SSRs — not-tested.

Would require a repeat-finder pipeline.

5 Critique of the replication

The REPLICATED verdict is real, but a serious reader should not stop reading here. See `report/failure_analysis.md` for the full failure/critique document. Summary:

- **Independence.** RefSeq NC_000907.1 is *derived from* the paper’s own 1995 GenBank submission L42023. It is not an independent measurement. So confirming that the paper’s numbers match RefSeq mostly certifies *lack of silent drift in the deposit*, not that the paper’s science has been independently validated.

The independent validation comes from three decades of subsequent *H. influenzae* Rd resequencing (which does corroborate the assembly) but that was not itself re-run in this replication.

- **Shortcuts.** No PDF (Unpaywall confirms no OA copy); no Marker/Nougat extraction; no re-BLAST for functional roles; no re-run of 1995-vintage ORF caller; no re-assembly of raw reads; no prophage/SSR re-detection with modern tools.
- **Hand-waved tolerances.** “Within re-annotation drift” is a folk-wisdom band ($\sim 5\%$), not a calibrated statistical threshold. A better replication would empirically calibrate drift by aggregating pre-2000 genome papers vs modern RefSeq.
- **LLM-judge is not independent.** The judge saw both sides. Its coverage=100 / agreement=100 output is a formatter of the numeric truth, not an oracle.
- **Method claim is unverified.** The paper’s headline methodological contribution — shotgun sequencing works on a free-living genome — is left explicitly untested. This is the paper’s most historically consequential claim.
- **Annotation loss.** The paper’s biological richness (prophages, contingency loci) is measurably *absent* from the modern reference record for this strain. The replication surfaces this problem but does not fix it.

The REPLICATED verdict should therefore be read as “**the deposited sequence faithfully carries the paper’s quantitative claims to within re-annotation drift**” and not as “**the paper’s science has been independently validated end-to-end.**” Both statements are true; only the first is what this replication established.

6 Verdict

REPLICATED.

Justification. 8/8 measurable quantitative claims (C1–C8, C13) reproduce against NC_000907.1 with deltas at the exact-integer level (length within ± 1 bp, rRNA operons/16S/23S loci exact, circular topology exact) or within the re-annotation drift band (CDS -1.3% , tRNA $+5.6\%$, mean CDS length $\pm 5\%$). G+C matches at 38.15% vs $\sim 38\%$. No measured quantity contradicts the paper. Not-tested items (C10, C11, C14) are historically or resource-limited, not evidence of a failed replication; they are declared out of scope in §5 and drive the open questions in §7. LLM-judge (Argo `argo:gpt-5`) corroborates with coverage_pct=100 / agreement_pct=100 but is treated as a structured summary of the numeric truth, not an independent oracle.

7 Open Questions

Five open questions grounded in what the paper leaves unresolved and what this replication surfaced. Each has concrete next steps. Full form in `report/open_questions.json`.

Q1. Can the original 1995 TIGR assembly be re-executed byte-for-byte?

Basis. The paper’s headline *method* claim (whole-genome random shotgun + TIGR Assembler produced a complete circular chromosome) is the most historically consequential contribution but our replication explicitly could NOT test it. The 1995 raw Sanger traces ($\sim 19,687$ reads) are not deposited in a standard SRA-reusable form; TIGR Assembler source has been superseded; the ~ 140 primer-walk gap-closure notebooks are text-only. This is the single largest untested piece of the Fleischmann corpus. **Next steps.** (1) FOIA/archival request to JCVI for DAT/DDS backups of the 1995 traces; (2) cross-reference NCBI Trace Archive mirror at EBI/ENA; (3) if reads recoverable, re-run SPAdes/Flye/Unicycler at matched $\sim 6\times$ coverage vs 1995 TIGR Assembler output; (4) if not, downsample modern Illumina KW20 data to $\sim 6\times$ and demonstrate 1990s-era-comparable input still closes; (5) write up as a ‘reproducibility of the first genome’ historical-methods paper.

Q2. What are the 117 pseudogenes in NC_000907.1 — real, artifact, or annotation choice?

Basis. Our replication reproduced CDS count within drift but did NOT audit the 117 pseudogenes. Are they genuine loss-of-function events in the KW20 lab strain, 1995-sequencing errors preserved in RefSeq, or artifacts of stricter modern calling? No published per-locus audit exists. **Next steps.** (1) Extract all 117 pseudo features + `note` qualifiers; (2) verify with $\geq 50\times$ short-read resequencing of fresh KW20 stock (SAMN02603729 or recent SRA); (3) classify functional-category distribution; (4) publish as a BV-BRC augmentation.

Q3. Which specific 3 tRNA loci were added between 1995 and RefSeq 2020?

Basis. +3 tRNA delta is flagged as “drift” but the specific loci and mechanism (missed by 1995 tRNAscan, sec/pyl-related, false-positive by modern tools, or comparative-genomics duplicates) are unknown. Systemic issue for all pre-2000 bacterial genome papers. **Next steps.** (1) Enumerate all 57 tRNAs from NC_000907.1; (2) re-annotate L42023 with tRNAscan-SE v2.0; (3) diff vs 1995 tRNA table; (4) classify each of the 3 delta loci; (5) submit re-annotation note to BV-BRC / RefSeq.

Q4. What is the correct community-adoptable definition of “coding density”?

Basis. Our replication reported 82.53% under one reasonable choice but the paper’s implicit definition would yield a different number. Coding density is quoted routinely as a bacterial-lifestyle proxy without a standard. **Next steps.** (1) Enumerate 16 combinations of inclusion criteria; (2) compute all on 20 curated reference genomes spanning lifestyles (E. coli MG1655, M. tuberculosis, M. genitalium, B. aphidicola, R. prowazekii, Synechocystis, etc.); (3) show which choices scramble the ranking; (4) release `bactcodedensity` PyPI package; (5) retroactively benchmark ~ 100 literature claims.

Q5. Where did the paper’s prophage / SSR annotations go, and is RefSeq silently losing biology?

Basis. NC_000907.1 has 3 `misc_feature` + 1 `repeat_region` vs the paper’s substantive discussion of two Mu-like prophages and multiple phase-variation SSRs. RefSeq re-annotation strips narrative annotation. The field’s “reference” for H. influenzae Rd is now measurably less informative than the 1995 paper. **Next steps.** (1) Recover the paper’s coordinates for the two Mu prophages + SSRs; (2) independently re-detect with PHASTER, VirSorter2, CheckV, Tantan, IMPERFECT; (3) cross-tabulate lost annotation; (4) submit a lossless GFF3 supplement to BV-BRC; (5) generalize the audit to a random sample of pre-2000 bacterial genome papers to quantify the systemic “RefSeq amnesia” problem.

Companion files

- `report/REPORT.md` — original narrative markdown report.
- `report/open_questions.json` — machine-readable open-question list.
- `report/workflow.md` — comprehensive workflow + tools + effort estimate.
- `report/artifacts_summary.md` — inventory of every artifact.
- `report/failure_analysis.md` — honest failure analysis + full critique.
- `report/evidence/computed.json` — deterministic numeric outputs.
- `report/evidence/llm_judge.json` — LLM-judge structured verdict.
- `work/analyze.py`, `judge.py`, `Hinf_Rd_NC_000907.1.gb` — code + input.
- `extraction/marker.md`, `extraction/nougat.mmd` — pending stubs (no OA PDF).

- `paper.pdf.MISSING.md` — Unpaywall provenance note.

Original replication: Ollie / Argus subagent, 2026-07-04 America/Chicago. Backfill to 8-artifact standard: 2026-07-05. Free endpoints only (NCBI E-utilities public tier; Argo LLM proxy 127.0.0.1:44497). No paid endpoints used at any stage.